

Assignment 5 -- Due 2/13/2025

- Due Feb 13 by 11:59pm
- Points 10
- Submitting a file upload
- Available Feb 7 at 3pm - Feb 13 at 11:59pm

This assignment was locked Feb 13 at 11:59pm.

Homework 5 -- Due 13 February

1. Consider a network with 2 inputs, 1 hidden unit, but also with DIRECT connections from the input units to the output. Show how such a network can conceivably compute XOR.
2. Use **bp3.m** to learn the task with 3 binary inputs that has target values of 0 for the inputs **000** and **111**, and target values of 1 for the inputs **001**, **010**, **011**, **100**, **101**, **110**. The network should have 3 input units, 2 hidden units, and 1 output unit. Use **forw3.m** to check the activities of the outputs after training. Show the hidden units on a 2 dimensional scatterplot -- color coded, if you can!
3. Repeat exercise 2 with 4 binary inputs --> target values of 0 for inputs **0000** and **1111** target 1 for all other (14) inputs.
4. Train a network on the **XOR** task. Plot the representations of the training set in the hidden unit space. Challenge: Create a contour plot of the output unit's response in the **stimulus space**.